



Purpose

We wanted to make a game inspired by the classic cartoon Tom and Jerry. For our project, we have a user controlled Tom robot which objective is to catch the pesky Jerry robot. Using a vision-based position tracking, wireless communication, state-based game logic, the system turns a chase sequence into a fun robot game!

System Overview

Gameplay

- The player uses a PS2 controller to move the Tom robot
- Tom can send slow commands when it faces Jerry robot
- Once Tom gets close to Jerry, Tom can freeze Jerry
- During the 15 second frozen duration, the player has to input a button sequence presented on the monitor
- If the player mis-inputs the sequence, a new sequence combination is given
- If the player is unable to input the correct sequence in 15 seconds, Jerry starts running away again!
- If the play inputs the correct sequence in time then Jerry is captured! You win!

Key Components

Game Console

- DE2
- Picamera and Raspberry Pi 3B model
- Monitor - Puzzle + robot location display
- Sends commands and data to robots
- Wireless Communication

Tom

- PS2 controller
- Range-based Freeze trigger
- Manual Movement
- Wireless Communication

Jerry

- Wall detection
- Autonomous movement
- LED status display
- Wireless Communication



Concepts

Functional Diagram

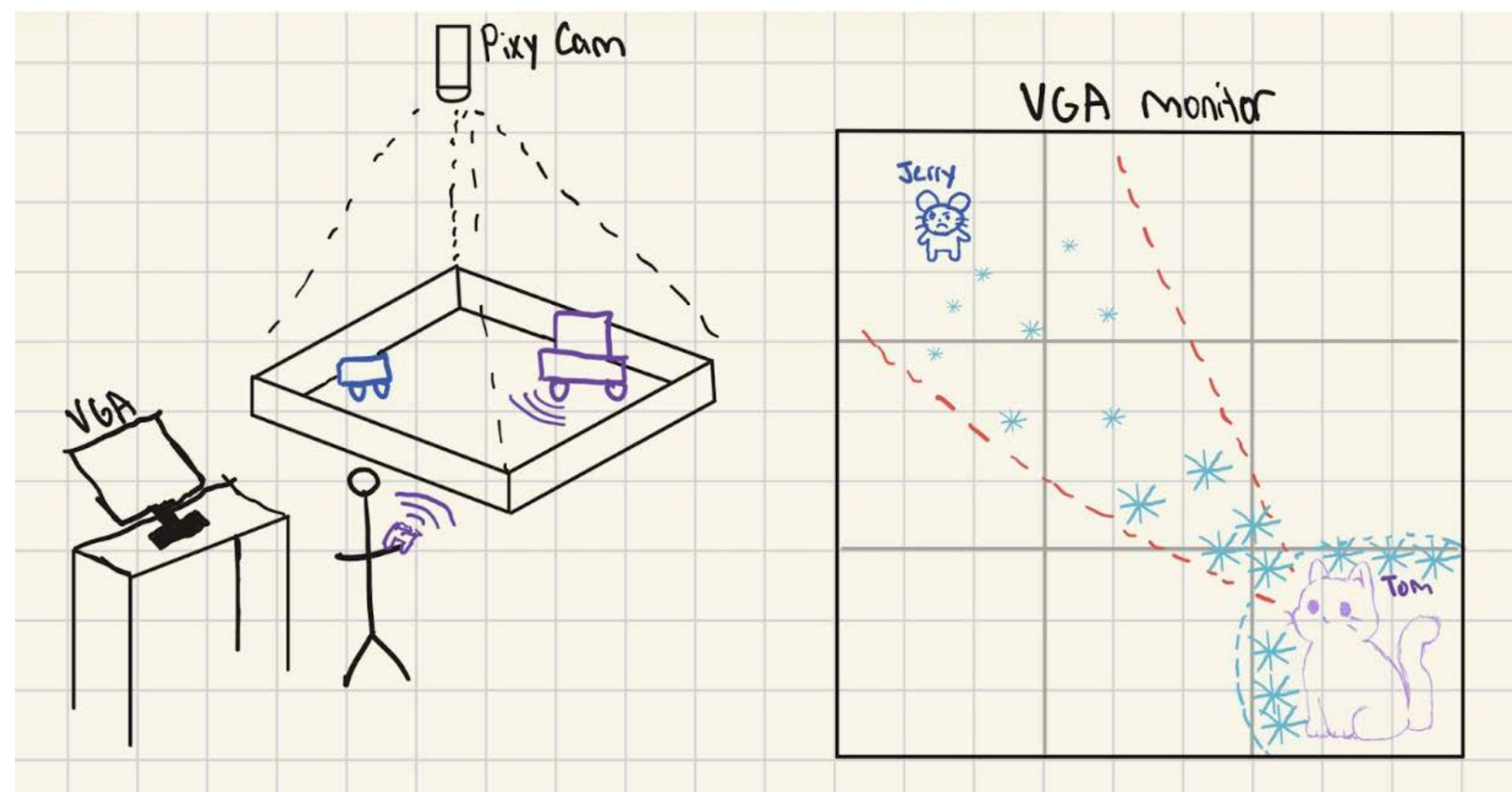


Fig 1. A functional diagram displaying the player, game console, Tom and Jerry. and the VGA monitor display

Communication Diagram

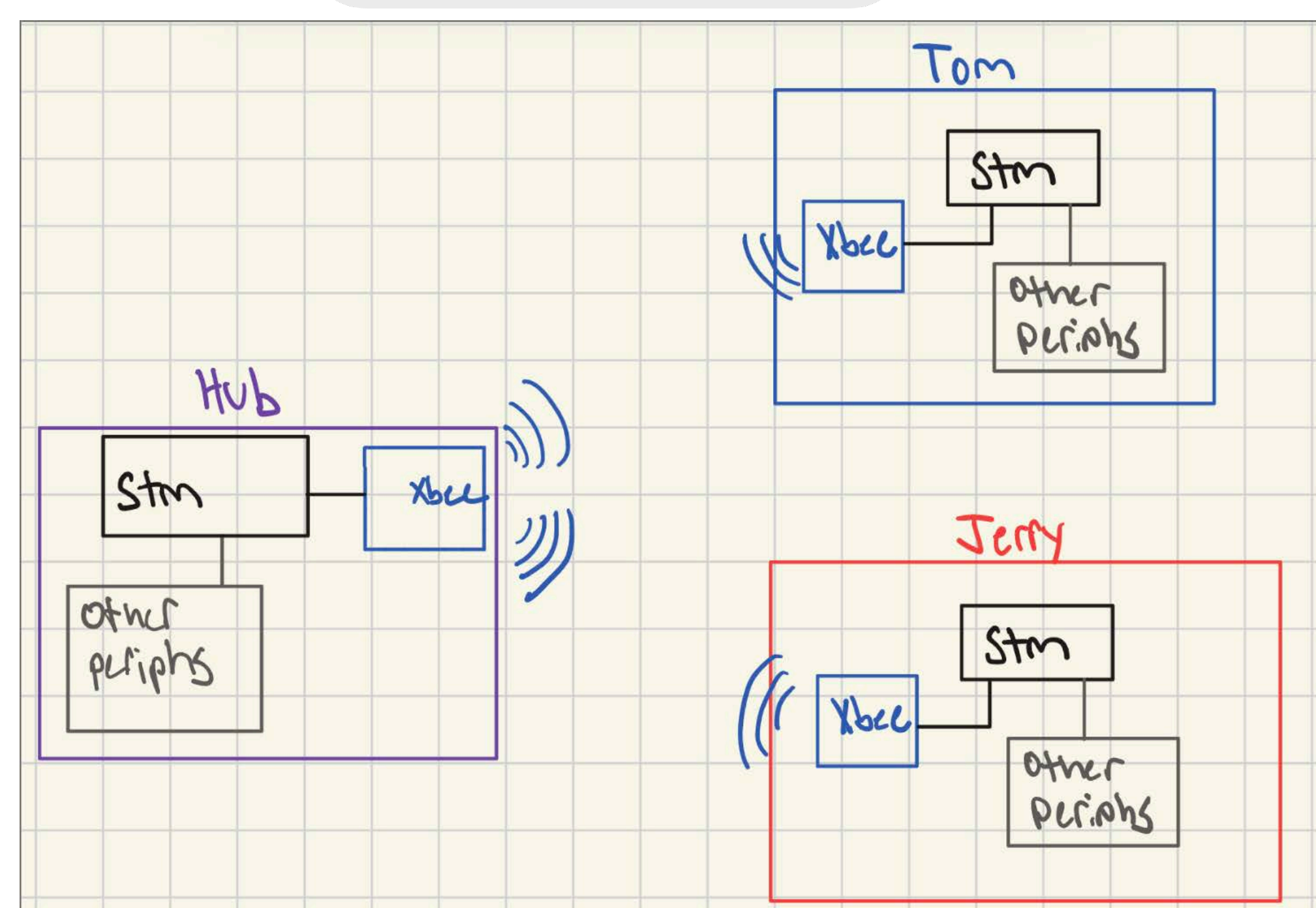


Fig 2. Star method of communication using 3 XBees. This allows noise filtering via changes

Results

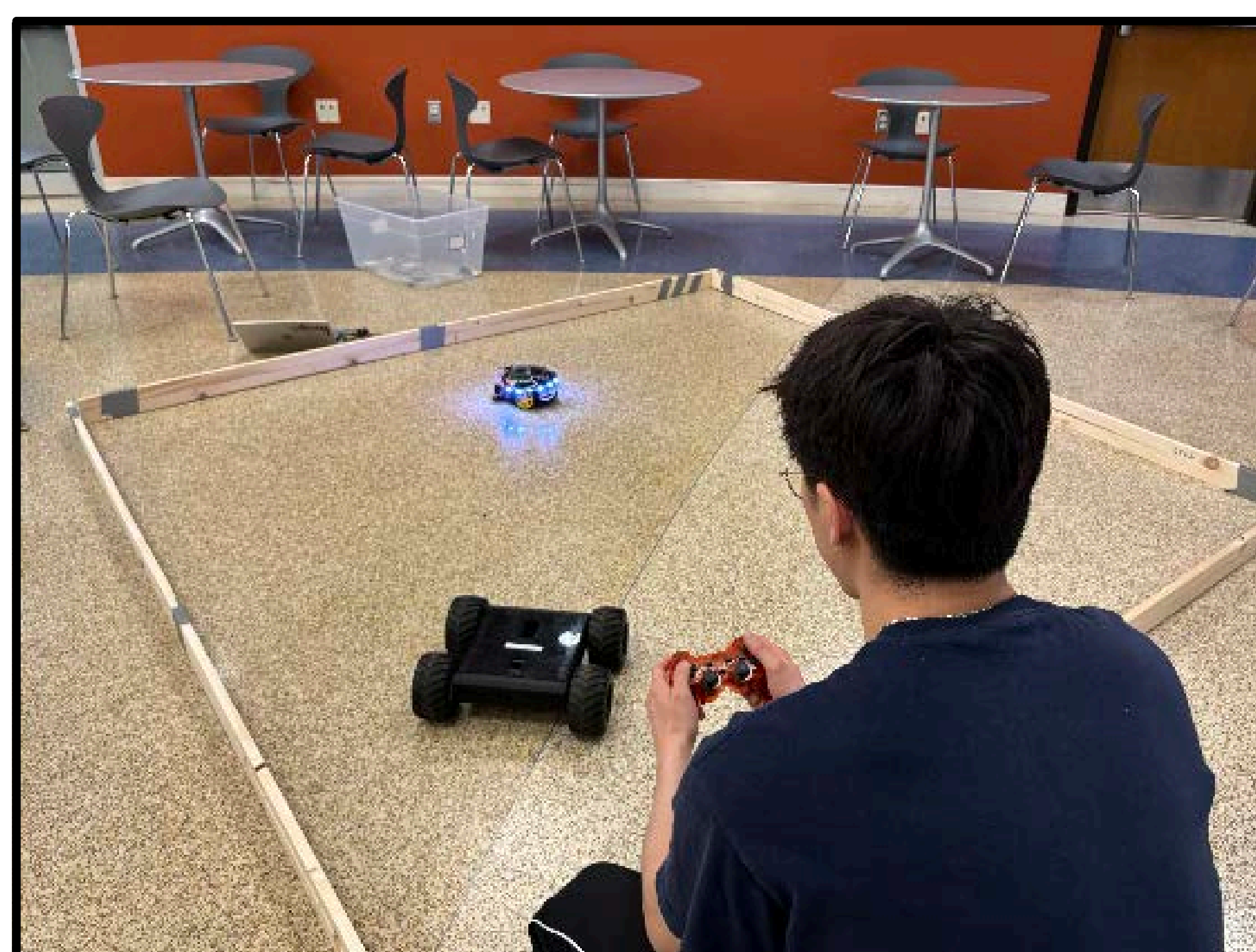


Fig 3. A player controlling Tom robot to catch and capture the Jerry robot

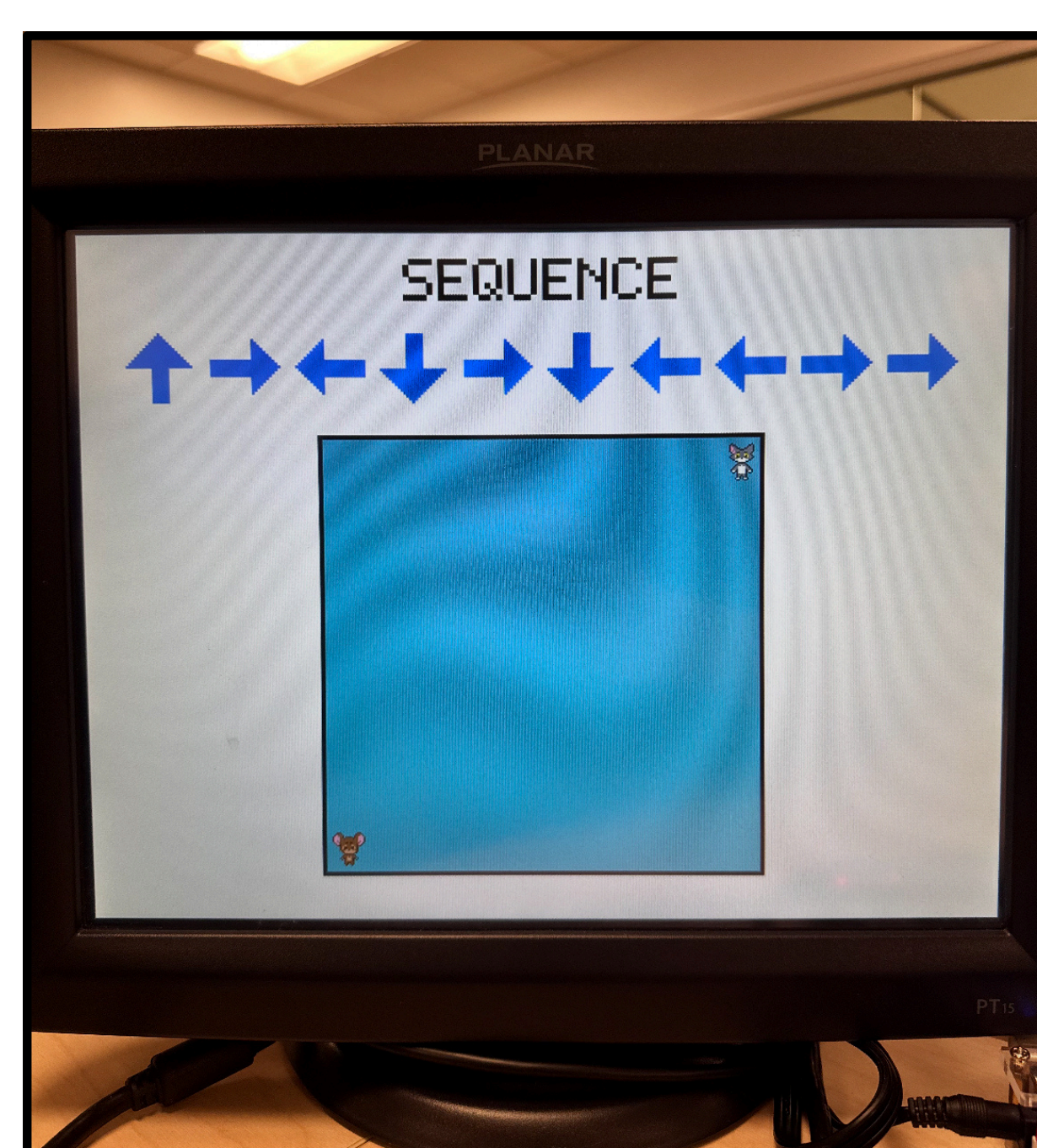


Fig 4. The monitor showing the winning sequence and a map of where Tom and Jerry are in the arena